Hrishikesh S Raj

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Professional Summary

Computer Science graduate focused on Machine Learning, Computer Vision, and Geospatial AI. Experienced in developing deep learning pipelines, backend systems, and retrieval-augmented AI applications using Python and PyTorch. Interested in building reliable AI systems and full stack applications through scalable software engineering.

Technical Skills

Programming: Python, C++, Java, SQL, JavaScript
Machine Learning: PyTorch, Scikit-learn, NumPy, Pandas, OpenCV
AI Techniques: Semantic Segmentation, Computer Vision, Evidential Deep Learning, RAG, Vector Databases
Backend: REST APIs, Flask, FastAPI
Geospatial: Rasterio, GDAL, DEM
Tools & Platforms: Git, GitHub, Linux, VS Code, LaTeX, Overleaf, Vercel, Render, QGIS

Publications

IEEE Conference Paper — *Accepted for Presentation | Date: August 21, 2026*
"Uncertainty-Aware SegFormer for Satellite Image Segmentation using EDL" — *First Author*
- Formulated an evidential deep learning approach using SegFormer to estimate pixel-wise uncertainty and out-of-distribution detection in high-resolution satellite raster datasets.

Projects

SegFormer: Uncertainty-Aware Satellite Image Segmentation — GitHub | *Jan 2026 – May 2026*
Technologies: Python, PyTorch, SegFormer, Rasterio, NumPy
- Developed a semantic segmentation pipeline using SegFormer for high-resolution multi-spectral satellite imagery.
- Integrated Evidential Deep Learning (EDL) for pixel-wise uncertainty estimation and failure localization.
- Built a Rasterio-based preprocessing pipeline to handle patch-based satellite tile streaming and data augmentation.
- Implemented custom evidential loss with KL divergence annealing and Out-of-Distribution (OOD) evaluation.
- Achieved 65.81% mIoU on the DeepGlobe dataset; documented findings in an IEEE paper accepted for presentation.

TerraWatch: Geospatial Landslide Susceptibility Mapping — GitHub | *Aug 2025 – Dec 2025*
Technologies: Python, Scikit-learn, Rasterio, NumPy, Matplotlib
- Preprocessed multi-spectral satellite rasters and Digital Elevation Models (DEM) using Rasterio to extract slope, aspect, and elevation features.
- Trained unsupervised K-Means clustering models in Scikit-learn to classify spatial terrain vectors into distinct landslide risk levels.
- Generated geospatial susceptibility heatmaps by mapping cluster outputs back to geo-referenced spatial coordinate grids.

QKD Simulator: Quantum Key Distribution Protocol — GitHub | *Jan 2025 – Mar 2025*
Technologies: Python, NumPy, Cryptography
- Simulated Quantum Key Distribution (QKD) protocols in Python to evaluate quantum-safe cryptographic handshakes.
- Implemented photon polarization state models and key generation algorithms to quantify error rates.
- Evaluated computational overhead and key strength characteristics against classical RSA and ECC algorithms.

Experience

Family Health Centre, Choolur — *Kozhikode, India*
Community Service Volunteer — *Short-term Engagement*
- Supported administrative workflows and digitized patient records to improve operational efficiency.
- Assisted medical personnel with patient queue management and routine operational data entry.
- Coordinated logistics and resource organization for community health awareness initiatives.

Education

SRM Institute of Science and Technology — *Chennai, India*
Bachelor of Technology in Computer Science and Engineering (CGPA: 7.41 / 10) — *Aug 2022 – May 2026*

Dayapuram Residential School — *Calicut, Kerala*
CBSE Class XII (CMPC) – 83.2% — *Graduated 2021*